

ABOUT

I'm a Mechanical Engineering graduate from Stony Brook University with experience across underwater robotics, aerospace systems, and computational fluid dynamics. My interests span the full stack in mobile systems engineering, always grounded in rigorous analysis and testing.

INSTITUTION	Stony Brook University
DEGREE	B.E. Mechanical Engineering, 2026
LOCATION	Long Island, NY

SKILLS

DESIGN & ANALYSIS	CAD/CAM · DFM/DFA · GD&T · ASME Y14.5 · Engineering Drawings · Basic FEA
FLUID DYNAMICS	OpenFOAM · CFD (RANS) · Paraview · Meshlab · HPC · Parallel Computing
CONTROL & ESTIMATION	MATLAB/Simulink · Control System Toolbox · ROS2 · Basic Sensor Fusion
FABRICATION & TEST	Additive Manufacturing · Subtractive Manufacturing · Basic Mechatronics · Systematic Test Procedures
PROGRAMMING	MATLAB · Python · Git · Linux/Unix

01 UNDERWATER SOFT ROBOTIC SYSTEM

Soft Flyers Group · Stony Brook University

ROBOTICS

January 2025 - Present

WHAT

- Full-stack architecture of an underwater robotic system
- Soft actuator design, chassis development, and sensor/electronics integration
- Investigating bio-inspired multimodal locomotion strategies

HOW

- Soft actuator design and dynamic characterization
- ROS2-based control software and sensor integration
- Systematic testing protocols across operating and transitioning media

RESULTS

- Novel characterization of actuator dynamics across fluid media
- Ongoing: full system integration from mechanical to software stack

02

COMPETITIVE LAUNCH VEHICLE

Stony Brook Student Launch Team · Stony Brook University

AEROSPACE

September 2025 - Present

WHAT

- Numerical characterization of launch vehicle flight envelope
- Mechanical design, fabrication, and integration of competitive subsystems

HOW

- Aerodynamic simulation and flight envelope analysis
- Subsystem mechanical design and fabrication
- Integration testing and ground verification

RESULTS

- Aerodynamic behavior characterized across full flight envelope
- Competitive subsystems fabricated, integrated, and verified

03

COMPRESSIBLE CFD: TRAPEZOIDAL FIN AERODYNAMICS

Advanced Fluid Dynamics & Propulsion Lab · Stony Brook University

RESEARCH

October 2025 - January 2026

WHAT

- Numerical investigation of trapezoidal fin aerodynamics in compressible flow
- Meshing support for Triply Periodic Minimal Surface heat exchangers

HOW

- Compressible RANS with Spalart-Allmaras turbulence closure
- OpenFOAM on HPC cluster with parallel computing
- Systematic geometric parameter sweep

RESULTS

- **30.7% drag reduction** identified at a 46% lift-to-drag ratio penalty
- Quantified aerodynamic trade-space for fin geometry optimization
- Advanced TPMS heat exchanger meshing methodology

04

AUTONOMOUS AEROBOT THERMAL CONTROL SYSTEM

NASA L'SPACE Mission Concept Academy · NASA L'Space

RESEARCH

January 2025 - May 2025

WHAT

- Thermal analysis for an autonomous planetary aerobot
- Passive thermal control system design following the NASA mission lifecycle

HOW

- Churchill-Bernstein convection models and heat flow mapping
- Variable geometry radiator with low emissivity coating ($\epsilon = 0.008$)
- Multi-objective optimization across mass, power, and cost constraints

RESULTS

- **30 kW cooling** and **11 kW heating** requirements quantified across 350–400 K atmospheric profile
- 4 kg passive thermal system meeting \$6.3M cost constraint
- Design validated against full NASA mission lifecycle requirements

05

SOLAR BOAT POWERTRAIN PACKAGING

Stony Brook Solar Boat Racing Team · Stony Brook University

DESIGN

September 2022 - November 2024

WHAT

- Powertrain packaging for a solar-powered competitive racing boat
- Integration and operational readiness for competitive conditions

HOW

- Additive manufacturing for structural packaging investigation
- Systematic wet testing and diagnostics for component validation
- Robustness-driven design methodology

RESULTS

- Operational readiness confirmed through systematic wet testing
- Robust packaging solution developed via additive manufacturing

06 ROCK VARNISH GEOCHEMICAL PROVENANCE ANALYSIS

BNL SPARK Program · Brookhaven National Laboratory

RESEARCH

September 2018 – May 2022

WHAT

- XRD and XRF investigation of Fe/Mn stratigraphic layering in rock varnish
- Provenance analysis of steatite archaeological artifacts

HOW

- X-ray diffraction (2.0–7.5 Å) and XRF elemental mapping
- d-spacing fingerprinting for material sourcing
- Comparative elemental analysis across 10+ artifact samples

RESULTS

- Fe/Mn stratigraphic layering identified in 6 samples — consistent with Younger Dryas formation
- 10+ steatite artifacts traced to Rhode Island quarry sources
- **10–45× elemental variation** distinguishing regional clay sources and trade patterns

07 FIRST ROBOTICS COMPETITION SYSTEMS

FIRST Robotics · Longwood High School

ROBOTICS

September 2018 – May 2022

WHAT

- Competitive robotic system design for FIRST Robotics competitions
- Multidisciplinary team collaboration through full engineering design process

HOW

- Iterative mechanical design and prototyping
- Cross-functional integration with electrical and software teams
- Systematic testing and competition preparation

RESULTS

- Competition-ready robotic systems designed and deployed
- Engineering design process applied through full project lifecycle